Vortex

OrderGuard

SDC HACKATHON 2026 · DOMAIN 2

Catch supplier discrepancies before they become expensive mistakes.

One explainable workflow that connects supplier messages, purchase orders, receiving records and invoices — while keeping the SME owner in control.

Check. Explain. Remember. Approve.

Live demo

sdc-hackathon-submission-template.replit.app

SCAN LIVE DEMO

Team Vortex

The SME problem

Scattered evidence creates manual cross-checking, repeated follow-up and inconsistent exception handling.

MSME contribution to GDP

39.7%

Malaysia · 2025

MSME share of employment

48.7%

Malaysia · 2025

The real friction

- Supplier terms live in messages, POs, receiving notes and invoices.
- Staff must compare quantity, unit and price across those records.
- Ambiguous units can turn a simple check into back-and-forth follow-up.
- Repeat issues are hard to remember when history is scattered.

MANUAL EXCEPTION PATH

Supplier msg
"10kg chicken RM13/kg"

Purchase order
10 kg · RM12/kg

Receiving
8 kg received

Invoice
10 kg · RM13/kg



MANUAL CROSS-CHECK

Did quantity match?
Was the agreed price used?
Are the units comparable?
Has this happened before?
Who can approve follow-up?

OrderGuard focuses on the exception — not on replacing the SME owner.

Sources: DOSM, MSMEs Performance 2025 (published 30 Jul 2026); World Bank, Digitalizing SMEs to Boost Competitiveness (2022).

One complete exception workflow

From trigger to logged outcome — with AI assisting extraction, not making the business decision.

1 **TRIGGER**
Supplier message / invoice

2 **CONNECT**
PO + Receiving + Invoice

3 **CHECK**
Deterministic rules

4 **REMEMBER**
Supplier Pattern Memory

5 **GATE**
Trust Gate

6 **DECIDE**
Human approval

7 **LOG**
Outbox + audit trail

Design principle

AI extracts suggestions. Rules compare the evidence. Trust Gate blocks unsafe action. The owner decides.

This is intentionally not a fully autonomous financial agent.

What makes OrderGuard different

A focused review layer for SMEs — not a new accounting method, and not a chatbot making financial decisions.

AI

AI where it helps

Turns informal supplier text into structured suggestions.

RULES

Deterministic checks

Quantity, unit and price differences come from transparent rules.

BLOCK

Trust Gate

Blocks approval when inputs are not safely comparable.

MEMORY

Pattern Memory

Surfaces recurring supplier + item exception types.

RM

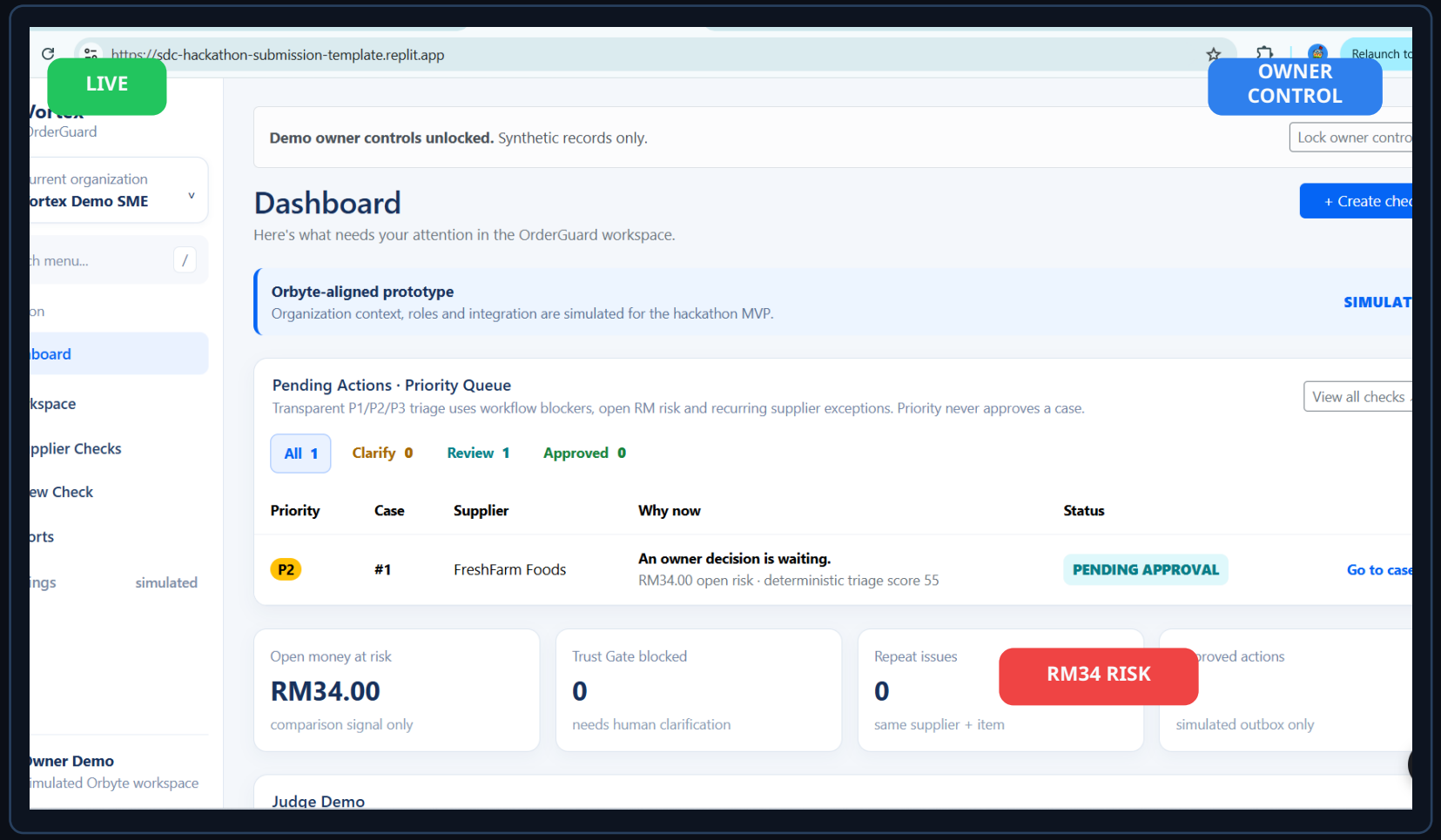
Money at Risk

Translates open discrepancy into an RM review signal.

Boundary: three-way matching already exists in major accounting systems; OrderGuard's innovation is the explainable SME exception-review experience and safety layer.

The product is live

Published Replit demo with a judge-ready scenario flow and visible exception intelligence.



Judge demo sequence

1. **Discrepancy**
   10 kg ordered · 8 kg received
   Invoice at RM13/kg
2. **Repeat issue**
   Same supplier + item
   Pattern Memory surfaces history
3. **Ambiguous unit**
   10 kg vs 2 bags
   Trust Gate blocks approval
4. **Clean match**
   30 trays match
   No unnecessary action

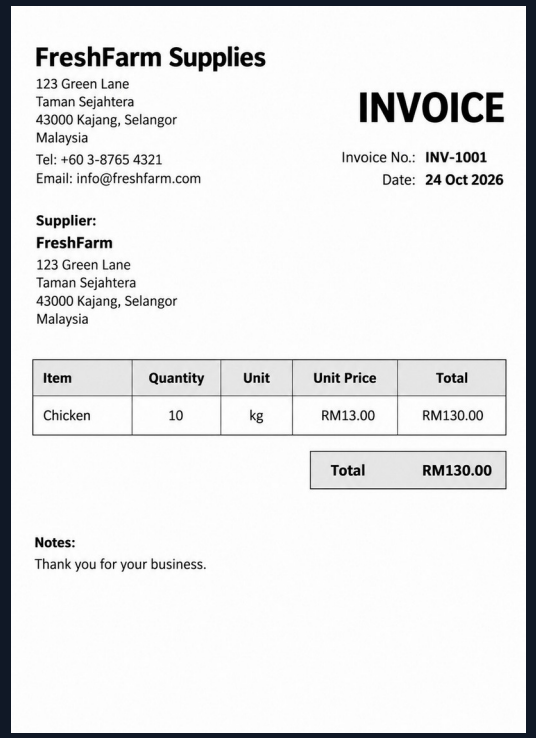
Live demo

sdc-hackathon-submission-template.replit.app

GPU OCR on RunPod

The invoice can be read on GPU infrastructure, but OCR never approves or rejects a case.

Synthetic invoice



FreshFarm Supplies
123 Green Lane
Taman Sejahtera
43000 Kajang, Selangor
Malaysia
Tel: +60 3-8765 4321
Email: info@freshfarm.com

INVOICE

Invoice No.: **INV-1001**
Date: **24 Oct 2026**

Supplier:
FreshFarm
123 Green Lane
Taman Sejahtera
43000 Kajang, Selangor
Malaysia

Item	Quantity	Unit	Unit Price	Total
Chicken	10	kg	RM13.00	RM130.00

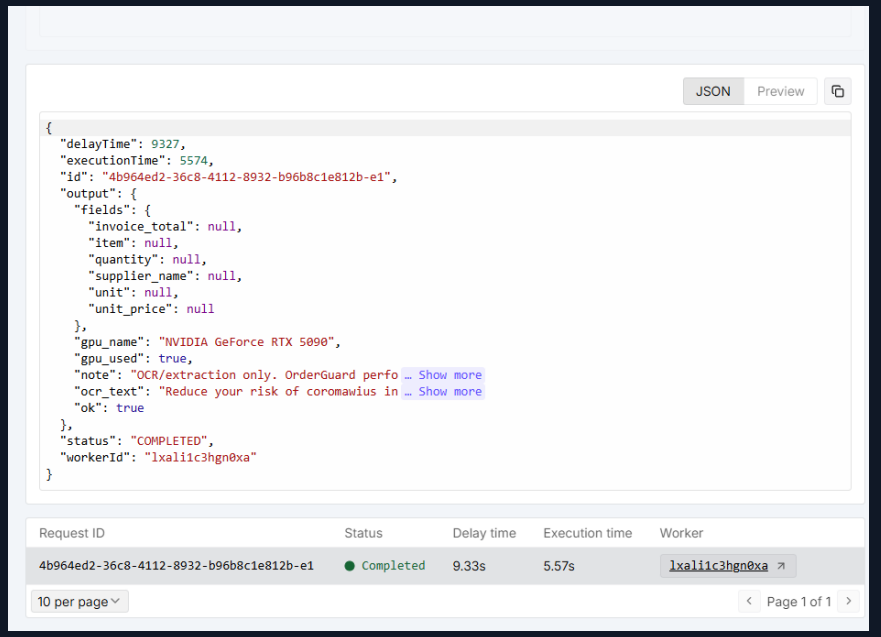
Total	**RM130.00**

Notes:
Thank you for your business.

FreshFarm · INV-1001 · 10 kg chicken · RM130

RUNPOD

Serverless GPU inference



COMPLETED | gpu_used: true | RTX 5090

What happens next?

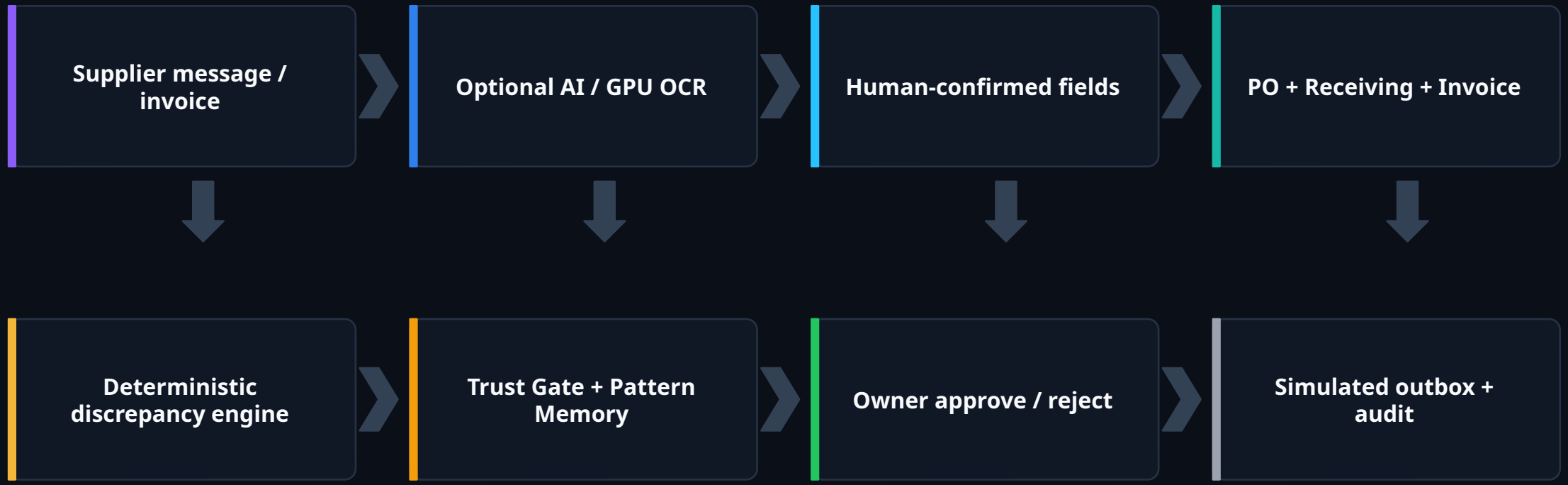
1. OCR text
2. Human-confirm fields
3. Deterministic compare
4. Trust Gate
5. Owner approval

No API key is exposed in the public repo or deck.

RunPod proof: request status COMPLETED, gpu_used=true, GPU reported as NVIDIA GeForce RTX 5090.

Architecture

Simple enough for an MVP, structured so the exception engine can become a reusable Orbyte module.



Prototype stack

Python · Flask · SQLite · Bootstrap · DeepSeek (optional) · RunPod OCR · Replit · GitHub Actions

No live Orbyte API integration is claimed in the MVP; Orbyte-facing roles and purchasing interfaces are simulated/aligned.

Human control is a feature

OrderGuard is designed to stop unsafe automation before it becomes a business action.

TRUST GATE

CLEAR	REVIEW	BLOCKED
Comparable records. No supplier action needed.	Discrepancy exists. Owner review required.	Ambiguous / incompatible. Human clarification first.

Example: 10 kg ≠ 2 bags

OrderGuard blocks approval instead of inventing a conversion.

AI OUTPUT

Must be human-confirmed

MUTATIONS

Owner access + CSRF

CONSEQUENTIAL ACTION

Owner approve / reject

DUPLICATE ACTION

Unique outbox protection

REJECTION

Creates no supplier action

DATA

Synthetic demo records only

Safety rationale aligns with NIST AI RMF GenAI Profile and OWASP guidance on output validation, limited permissions and human approval.

Built to fit Orbyte

A reusable purchasing-exception module, without pretending the MVP has a live Orbyte API integration.

POTENTIAL ORBYTE INPUTS

- Purchase orders
- Supplier records
- Receiving records
- Invoice data
- User / role context



ORDERGUARD

Purchasing exception engine

Compare · Explain · Gate · Remember · Approve



POTENTIAL OUTPUTS

- Exception case
- Approval / rejection
- Clarification request
- Audit / activity event
- Supplier risk context

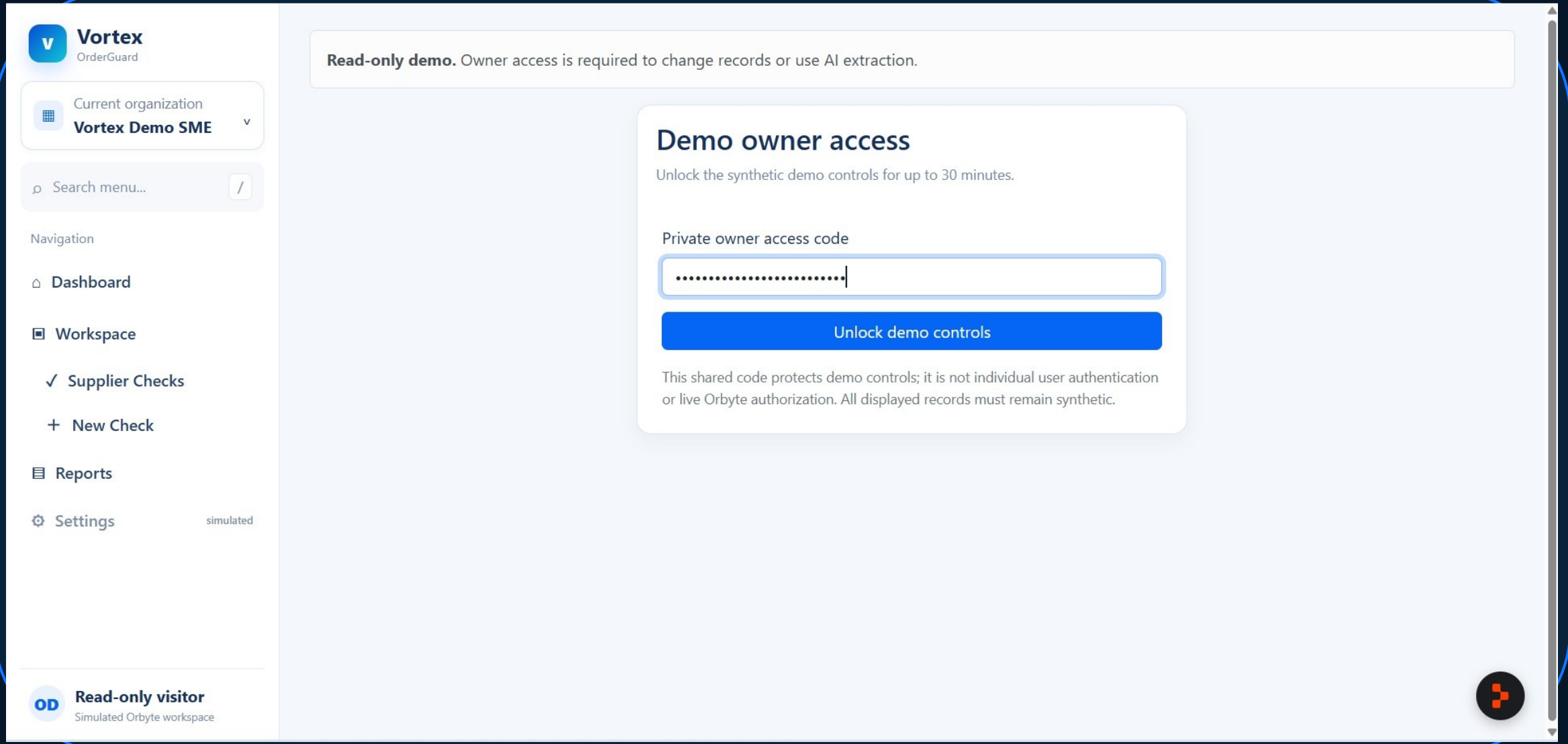
Reusable beyond one restaurant

Restaurants · retailers · small wholesalers · other SMEs

Demo owner access

Unlock the synthetic owner controls before the live walkthrough.

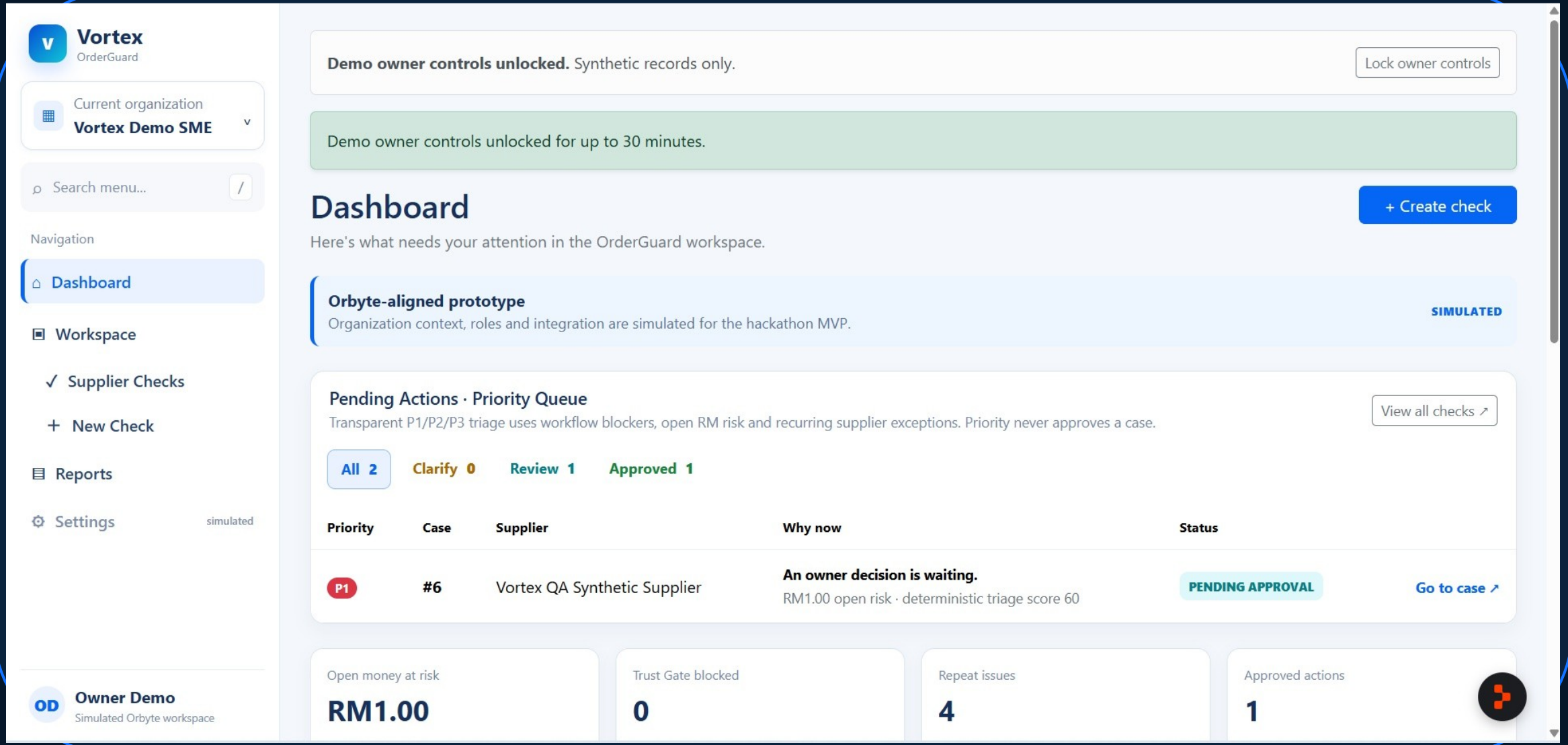
LIVE · 01



Dashboard overview

The review queue, open risk and repeat signals are visible in one place.

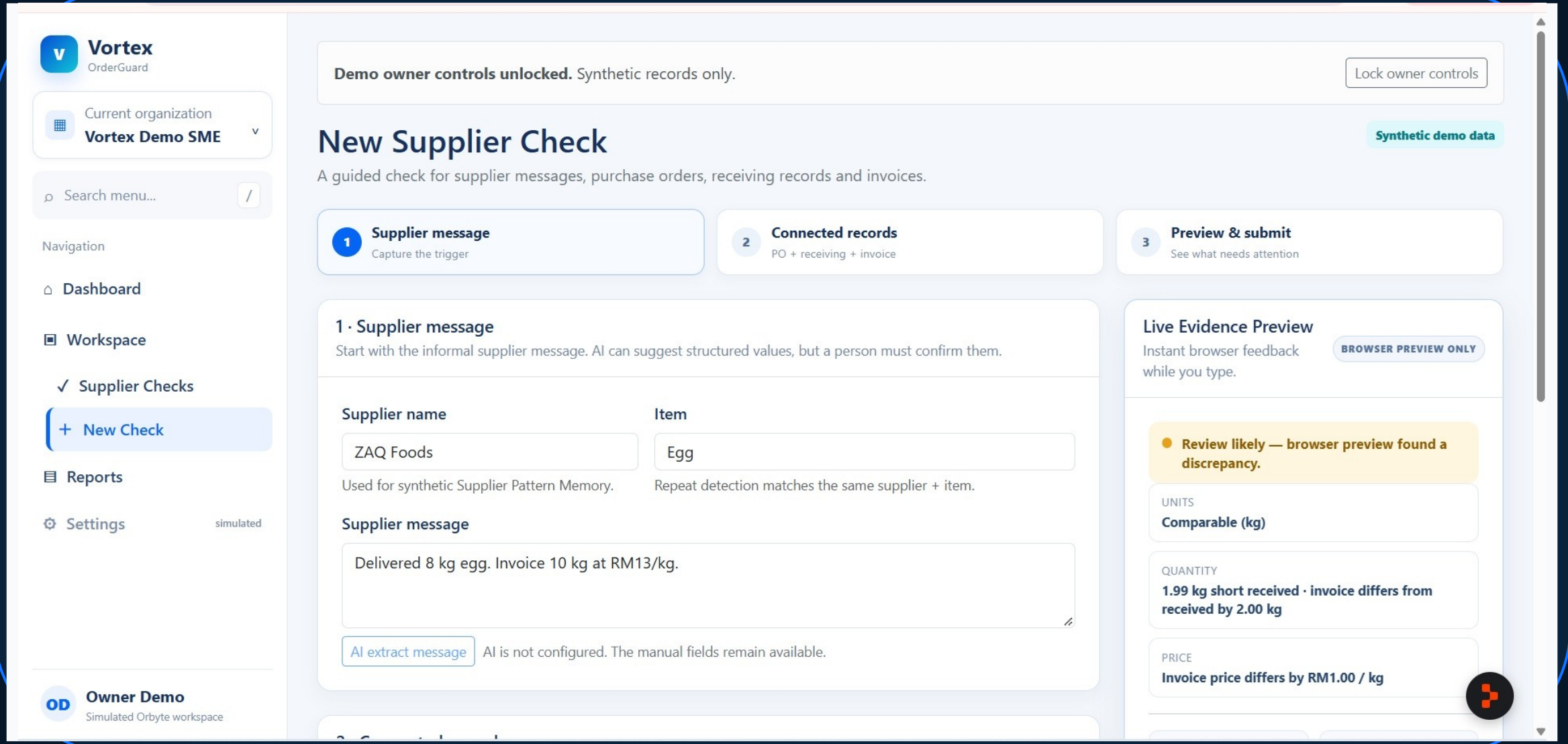
LIVE · 02



New check — supplier message

Capture the supplier trigger and preview discrepancies before submission.

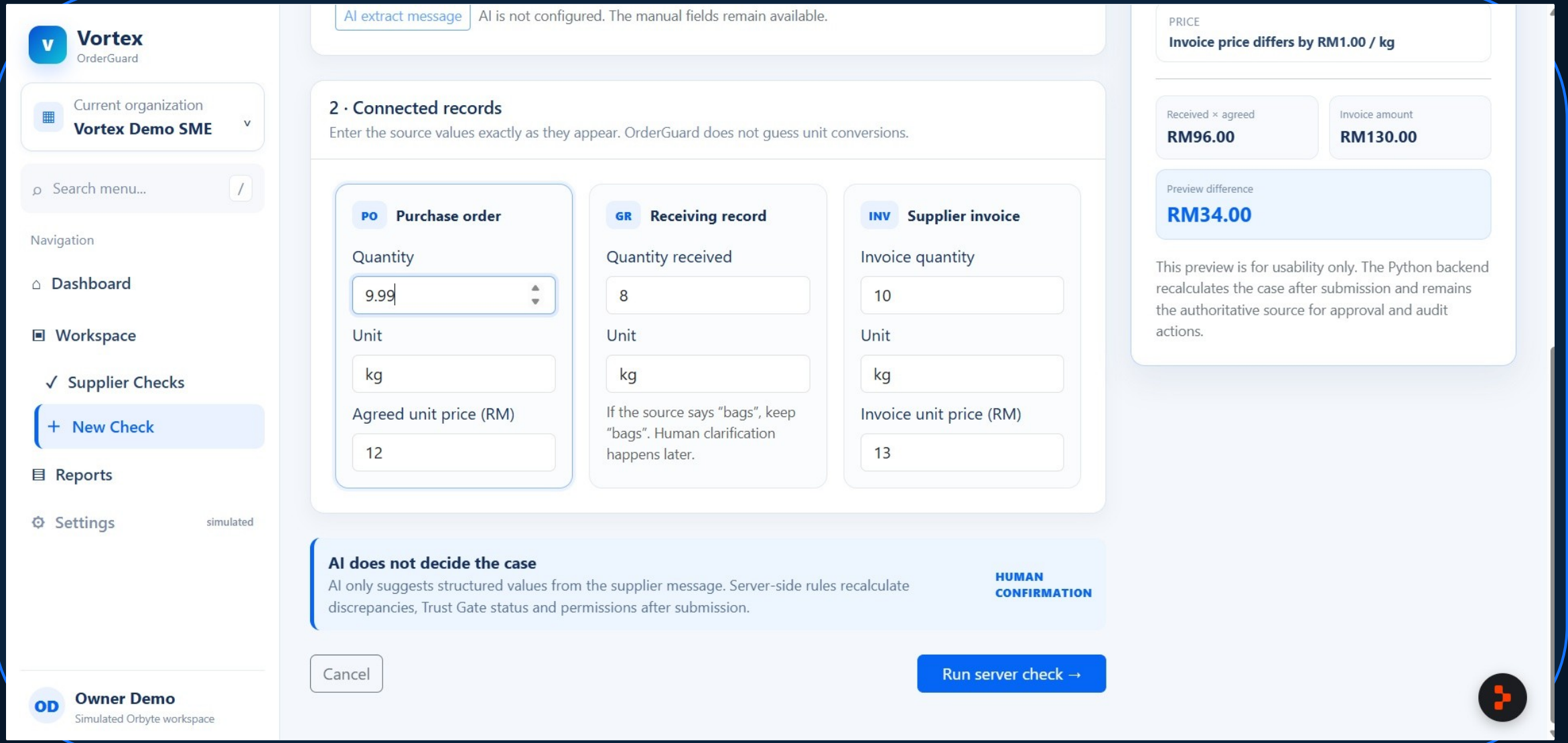
LIVE · 03



New check — connected records

Compare purchase order, receiving record and supplier invoice side by side.

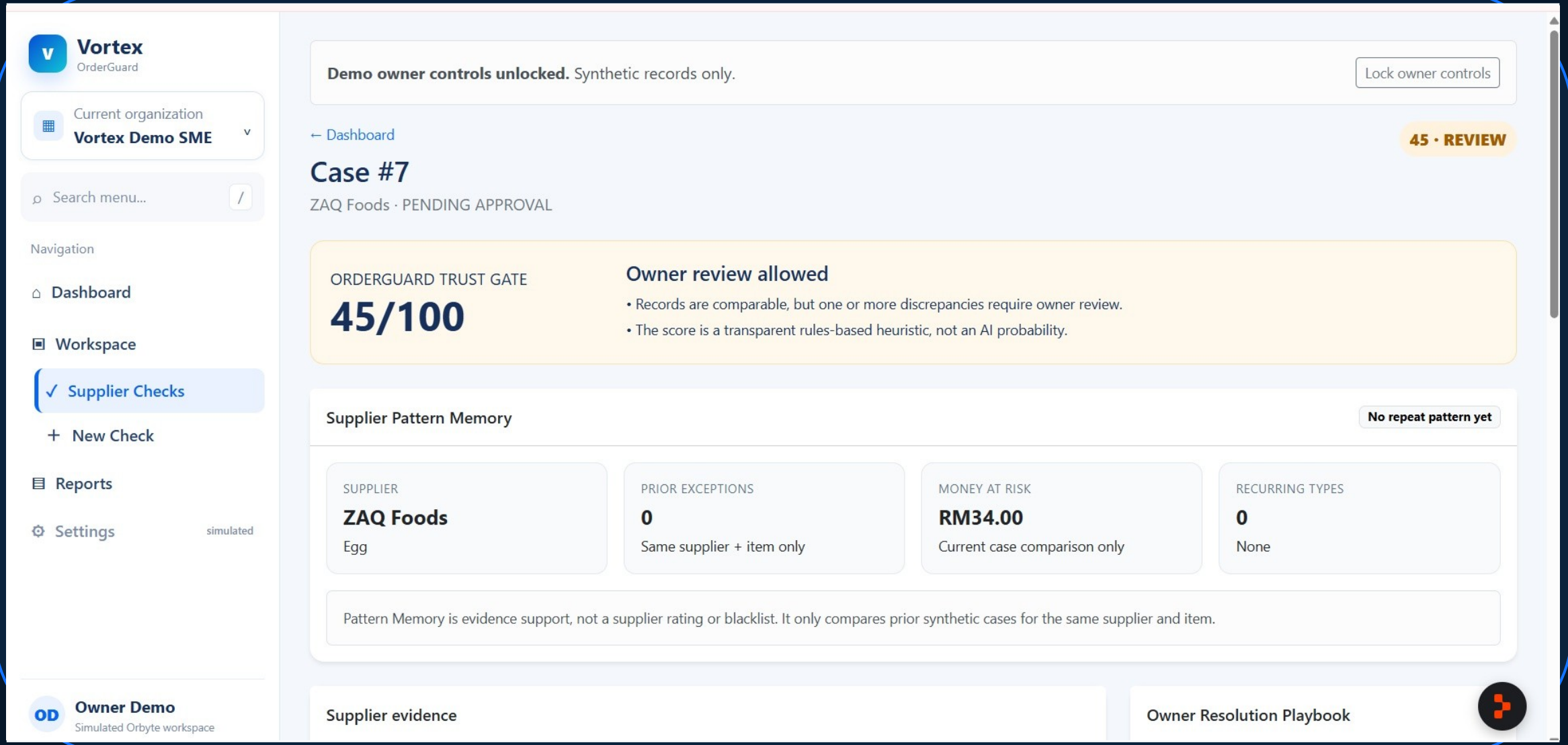
LIVE · 04



Case review — Trust Gate

Rules surface the review state while the owner remains in control.

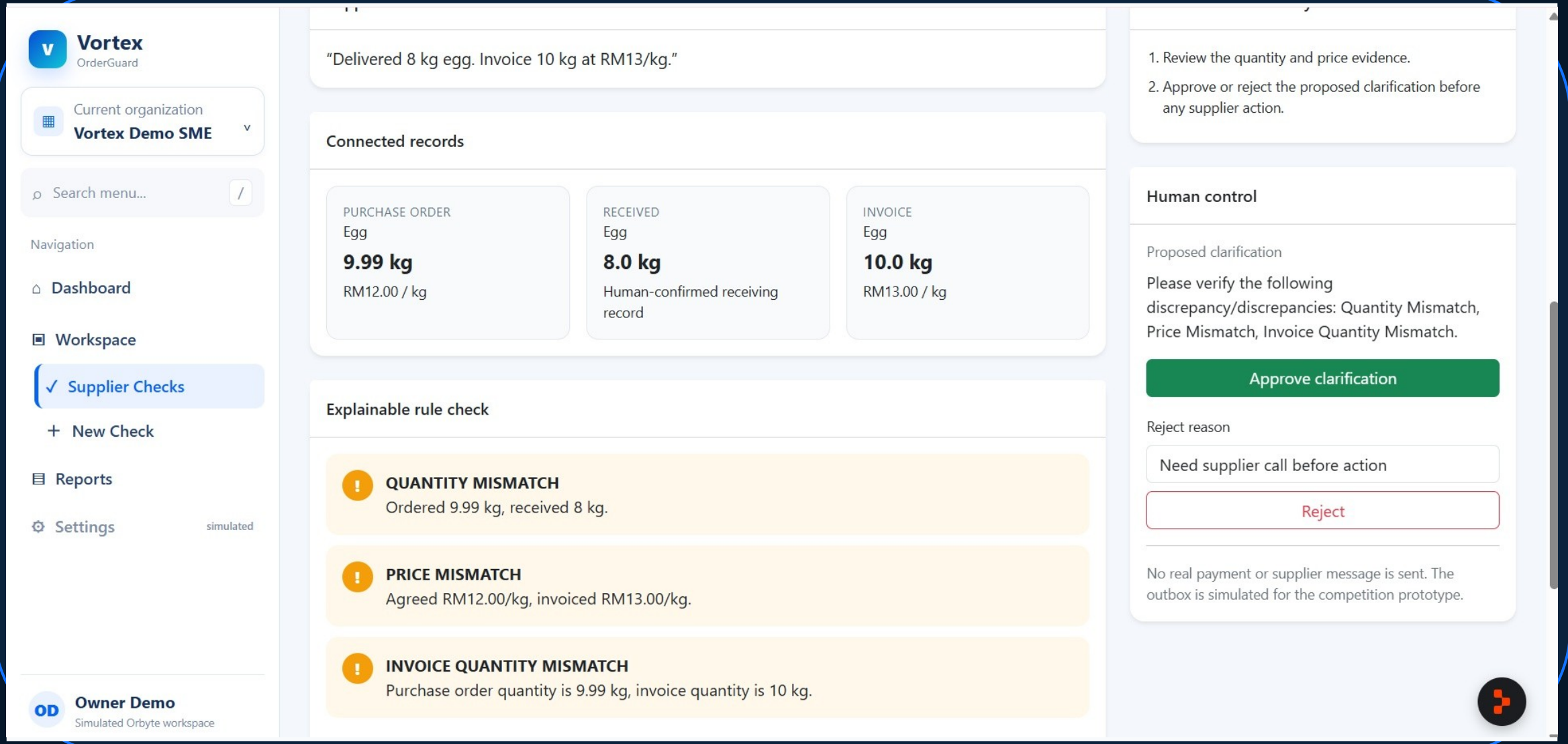
LIVE · 05



Explainable discrepancies

Quantity, price and invoice differences are shown with a human decision path.

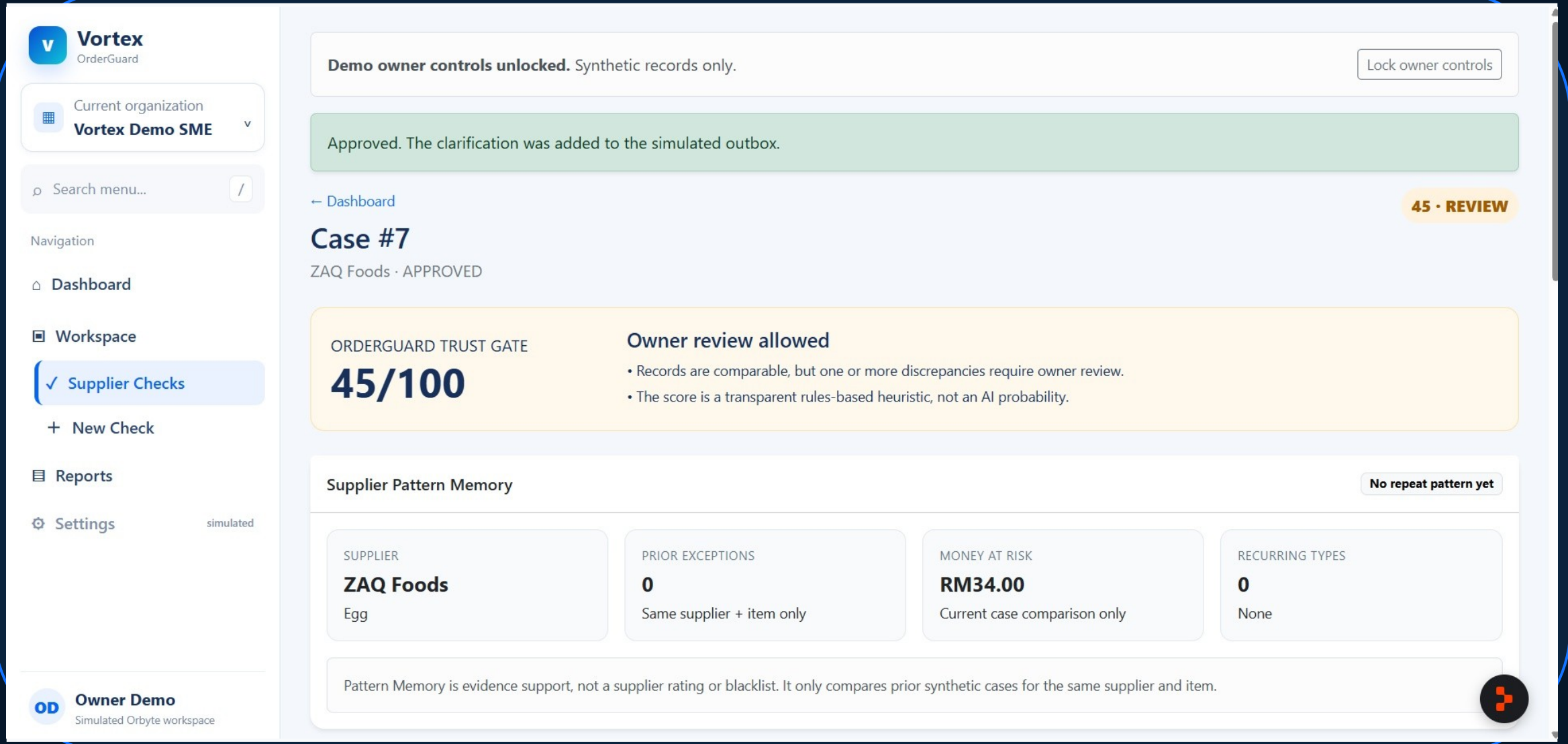
LIVE · 06



Approved outcome

The approved clarification is recorded in the simulated outbox.

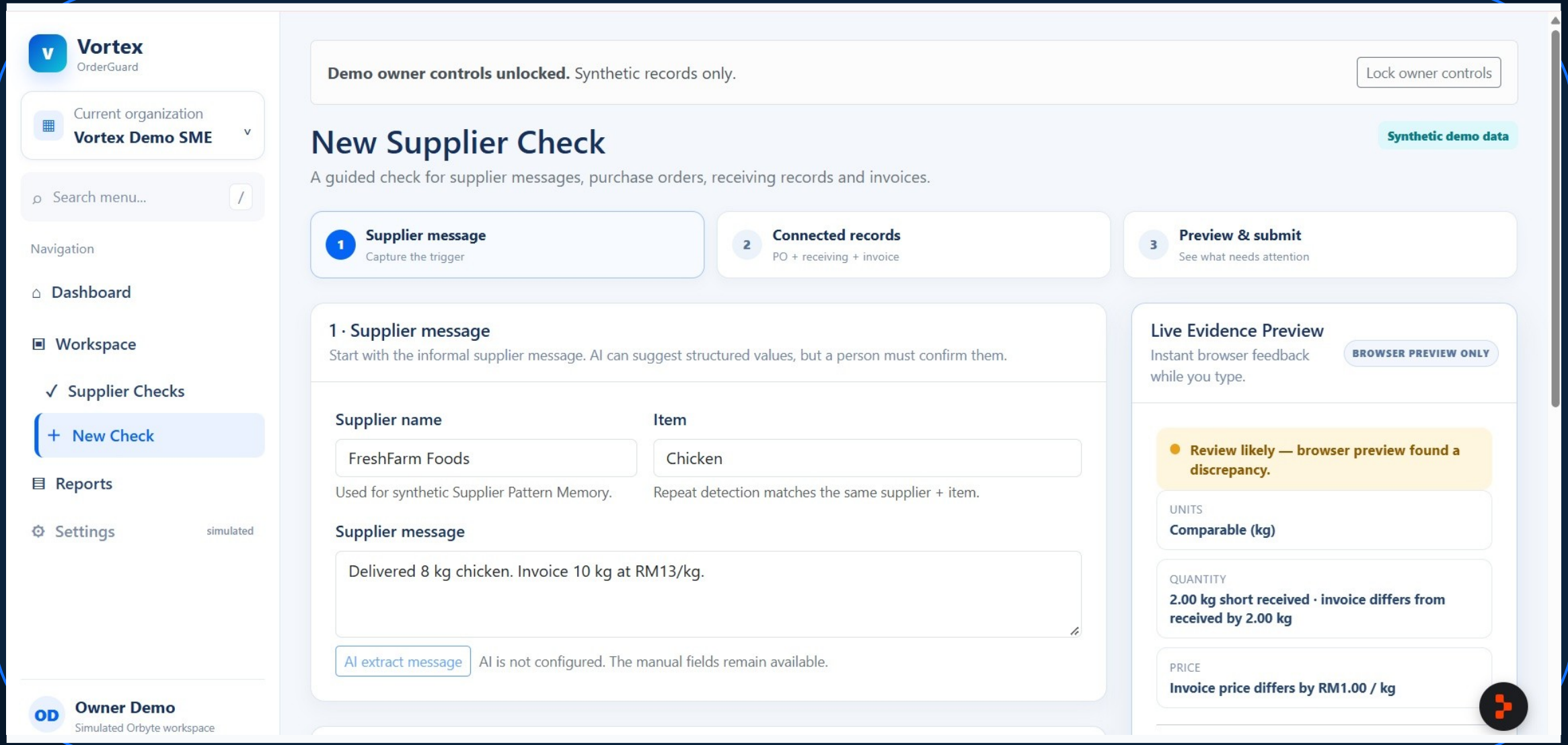
LIVE · 07



FreshFarm scenario

A second synthetic example demonstrates the live evidence preview.

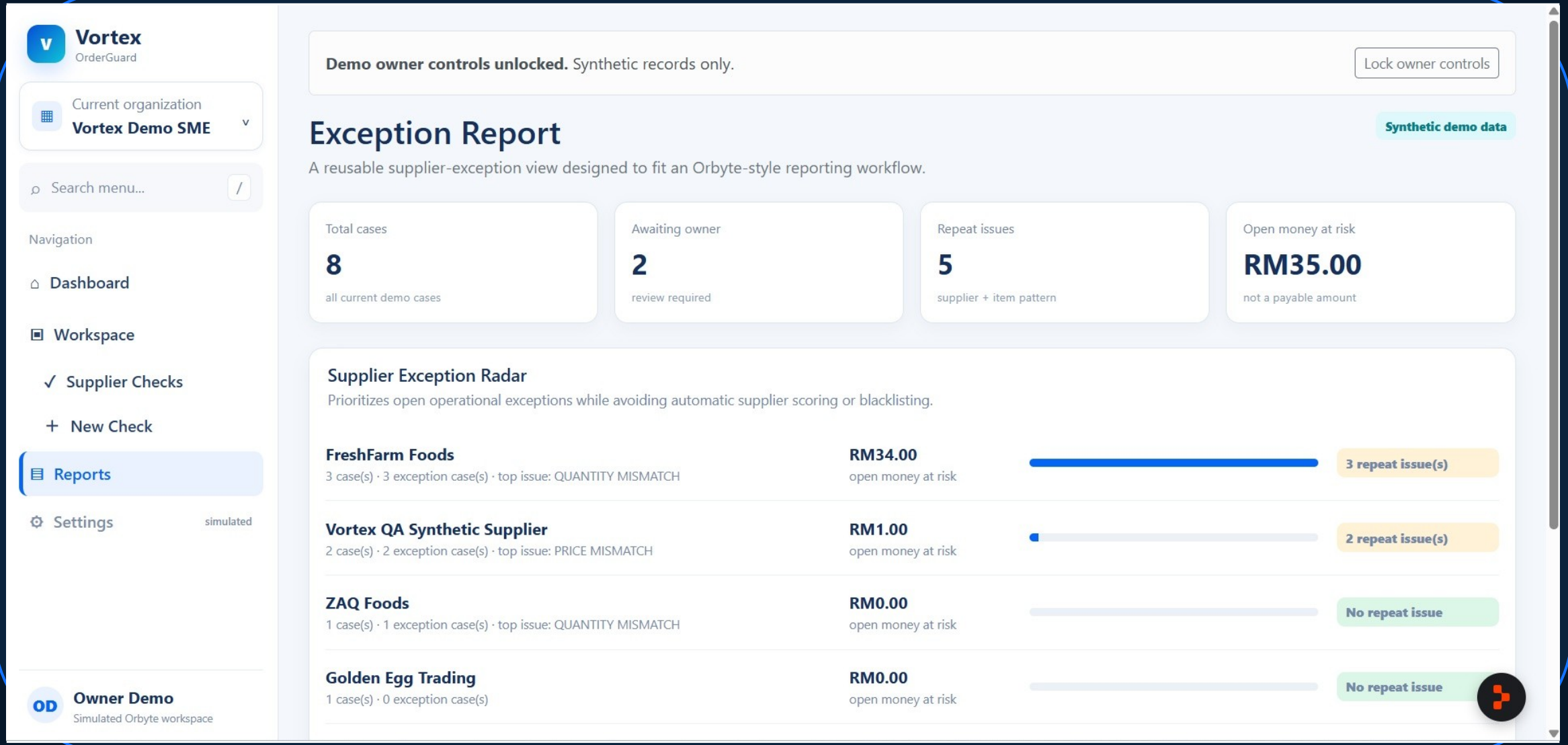
LIVE · 08



Exception report

Recurring supplier issues and money at risk are summarized for review.

LIVE · 09



Impact we can measure

We will not invent a savings percentage. The prototype is ready for a simple, defensible SME workflow test.

PRIMARY KPI

Average time to check one supplier case

Manual workflow vs OrderGuard workflow

- Use equivalent labeled synthetic cases.
- Include human review in both timings.
- Retain failed attempts and false alerts.
- Report sample size, errors and actual time difference.

Secondary checks

1	**Discrepancy detection**	Did known mismatches surface?
2	**False alerts**	Did clean matches stay quiet?
3	**Control tests**	Could anyone bypass the Trust Gate?
4	**Learnability**	Can a new tester complete the flow?

SDG 8 alignment: productivity through technological upgrading and MSME growth.

SDG alignment is intended alignment, not measured SDG impact. Source: United Nations SDG 8 targets 8.2 and 8.3.

V

Small businesses do not need another chatbot that tells them what might be wrong.

They need a reliable workflow that connects the evidence, explains the exception, remembers repeated issues — and still lets the owner make the final decision.

Vortex OrderGuard

Check. Explain. Remember. Approve.

TRY THE LIVE MVP

Live Demo

sdc-hackathon-submission-template.replit.app

GitHub

RunPod POU

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